American International University – Bangladesh
Faculty of Engineering
Department of Electrical and Electronic Engineering

Final Assignment			
Course Name:	Microprocessor and Embedded Systems with Lab	**Course Code:**	EEE 4103
Semester:	Fall 2025-2026	**Section:**	O
Faculty Name:	Dr. Md. Rukonuzzaman		
Submission Date:	27/12/25	**Due Date:**	**28/12/2025**

Course Outcome Mapping with Questions

Items	CO	POI	K	P	A	Marks	Obtained Marks
Q1-Q10	CO3	P.a.4.C.3	K4	P1, P3, P7		10	

Student Information:

Student Name:	Chinmoy Guha												**Section:**	O
Student ID #:	2	2	-	4	8	0	5	6	-	2	**Assigned Date:**	27/12/25	**Department:**	CSE
	p	q	-	a	b	c	d	e	-	r				

Assessment Rubrics:

COs-POIs	Excellent [10]	Proficient [8-9]	Good [6-7]	Acceptable [4-5]	Unacceptable [1-3]	No Response [0]	Secured Marks
CO3 P.a.4.C.3	All the problems are solved correctly. The simulation processes are clearly described, and results are generated by combining all possible input patterns with appropriate outcomes. All necessary drawings and computations are shown correctly.	All the problems are solved correctly. The simulation processes are clearly described, and results are generated by combining all possible input patterns with appropriate outcomes. A few necessary drawings and computations are missing but no wrong drawing.	All the problems are solved correctly. The simulation processes are not clearly described, and results are generated by combining all possible input patterns with appropriate outcomes. A few necessary drawings and computations are missing or wrong.	All the problems are not solved correctly. The simulation processes are not clearly described, and results are generated by combining several wrong or less no of input patterns with in/appropriate outcomes. Some necessary drawings and computations are missing or wrong.	All the problems are not solved correctly. The simulation processes are not described, and results are generated by combining mostly wrong input patterns with inappropriate outcomes. Almost all the necessary drawings and computations are missing or wrong.	No responses at all or copied from others	
Comments					**Total Marks (10)**		

1. Compute the duty cycle and sketch the waveform obtained at port D of the Arduino. Identify the modes of operation and compute the operating frequency of that mode based on the following program segment. Identify the Timer of the Arduino Microcontroller. The system clock frequency is (pq + r – b) MHz. Illustrate the relevant timing diagrams.

```
DDRD |= (1<<PD5);
pinMode(5, OUTPUT);
OCR0A = (200+ d + e);
OCR0B= (100 + a + b + c);
TCCR0A |= (1 << COM0B1) | (1 << COM0A0) | (1<<WGM01) | (1<<WGM00); TCCR0B |=
(1<<WGM02) | (1<<CS01) | (1<<CS00);
```

Name: Nafiur Rahman Siam

ID → 23-50834-1

Ans: No-1

1a) . Given,

pq-abcde-r

=23-50834-1

System clock frequency:

$f_{clk} = (pq+r-b)$ MHz

$=(23+1-0)$

$= 24$MHz

Identify the Timer Used:

Registers used:

OCR0A, OCR0B

TCCR0A, TCCR0B

Timer Used: Timer 0

Mode of operation (WGM bits)

$WGM02:0 = 111_2 = 7$

Mode 7 of Timer0:

Fast PWM mode with OCR0A as Top

Compare output modes:

OC0A (PD6)

→ COM0A0 = 1

COM0A1 = 0

OC0B (PD5)

→ COM0B1 = 1

COM0B0 = 0

Since PD5 is configured as output:

PWM output appears on PD5 (OC0B)

Timer Prescaler

CS01 = 1

CS00 = 1

CS02 = 0

Prescaler value:

N = 64

Date : ....

OCR Values:

Top value: $OCR0A = 200 + d + e = 200 + 3 + 4$
$= 207$

Compare value: $OCR0B = 100 + a + b + c = 100 + 5 + 0 + 8$
$= 113$

Operating frequency of PWM:

For Fast PWM with OCR0A as TOP:

$$f_{PWM} = \frac{f_{clk}}{N \times (OCR0A + 1)}$$

$$= \frac{24 \times 10^6}{64 \times (207+1)} = \frac{24 \times 10^6}{13312} \approx 1.80\ kHz$$

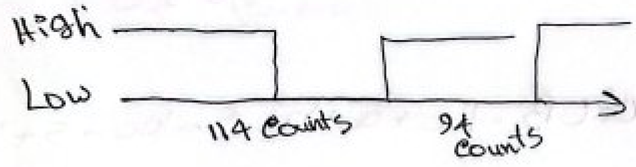
Duty cycle Calculation:

$$\text{Duty cycle} = \frac{OCR0B + 1}{OCR0A + 1} \times 100\%$$

$$= \frac{113+1}{207+1} \times 100 = \frac{114}{208} \times 100 \approx 54.8\%$$

9)

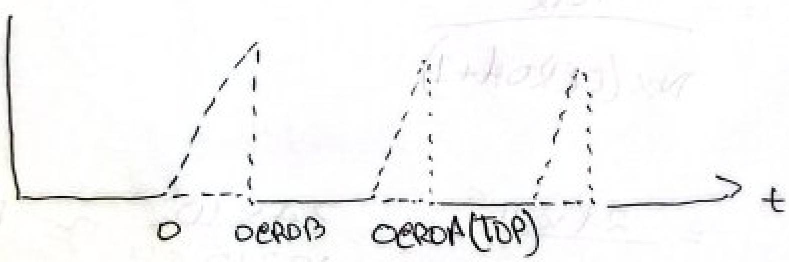
Waveform at PortD (PD5)



High duration = 114 timer counts

Low " = 208 - 114 = 94 timer counts

Timing diagram (fast PWM - mode 7)



Timer counts from $0 \rightarrow OCR0A$

Clears to 0 after reaching TOP

2. Calculate and identify the followings from the lines of codes mentioned in this question, Here the system clock frequency is (pq + r – b) MHz.
   a) System clock frequency
   b) Duty cycle
   c) Frequency of the PWM waveform
   d) Time period of the PWM waveform
   e) Waveform generation mode
   f) Compare output mode
   g) Waveform of the counter value
   h) Waveform of the output channel

```
DDRD |= (1<<PD6);
pinMode (6, OUTPUT);
OCR0A= (75 + a + b + e);
TCCR0A |= (1 << COM0A1) | (1<<WGM00);
TCCR0B |= (1<<CS00);
```

Solution:

Name - Suvra Chakraborty
ID - 22-48067-2
Course - Microprocessor and Embedded System
Sec - O

pq - abcde - r
22 - 48067 - 2

a. System clock frequency

$f_{clk} = (2R + 2 - 8)$

$PR = 2 \times 2 = 4$

$f_{clk} = (4 + 2 - 8) = -2MHz$

A negative clock frequency is not physically possible, the system will used by Atmega328P (Arduino uno)

is, $f_{clk} = 16MHz$

b. Duty cycle

calculation of OCR0A:

$OCR0A = 75 + a + b + e$
$= 75 + 4 + 8 + 7 = 94$

For 8-bit PWM

TOP = 255

Duty cycle $= \frac{OCR0A}{255} \times 100\%$

$= \frac{94}{255} \times 100 = 36.86\%$

Duty cycle $\approx 36.9\%$

c. Frequency of the PWM waveform

From control register

$CS00 = 1 \rightarrow$ Prescaler $N = 1$

PWM frequency formula (8-bit PWM)

$$f_{PWM} = \frac{f_{clk}}{N \times 256}$$

$$= \frac{16{,}000{,}000}{1 \times 256}$$

$$f_{PWM} = 62.5\ KHz$$

d. Time Period of the PWM waveform

$$T = \frac{1}{f_{PWM}}$$

$$= \frac{1}{62{,}500} = 16us$$

e. waveform generation mode

From code

$WGM00 = 1$

$WGM01 = 0$

$WGM02 = 0$

~~WA~~ WGM bits = 001

According Timer 0 mode selection

Phase correct PWM (8-bit)

f. Compare output mode

given code

COM0A1 = 1

COM0A0 = 0

This configuration corresponds to

None-inverting compare output mode

operation:

- output is set at Buttom
- output is cleared on compare match

g. waveform of the counter value

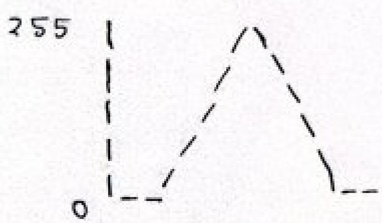
In phase correct PWM, the timer counter counts

from 0 to 255

Then 255 back to 0

Diagram

counter value



Timer ->0

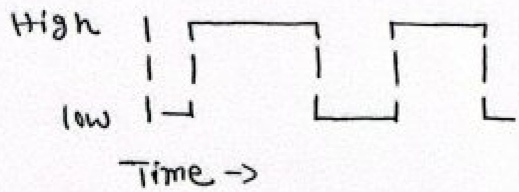
up-down (Triangular) counting

h. wave form of the output channel
(PD6 / OC0A)

- output Pin: PD6 (OC0A)
- Mode: Phase Correct PWM
- compare mode: Non-Inverting
- Duty cycle = 36.9%
- Frequency = 62.5 kHz

output waveform diagram

OC0A output



PWM wavform at PD6

Conclusion

The given code configures Timer0 of Atmega328P to generate a Phase correct PWM signal at Pin PD6 (OC0A) with a frequency of 62.5 kHz and a duty cycle of approximately 36.9%. using a Prescaler of 1 and non inverting compare output mode.

3. Design an a-bit shifter for the following shifting operations listed in the following Table 2:

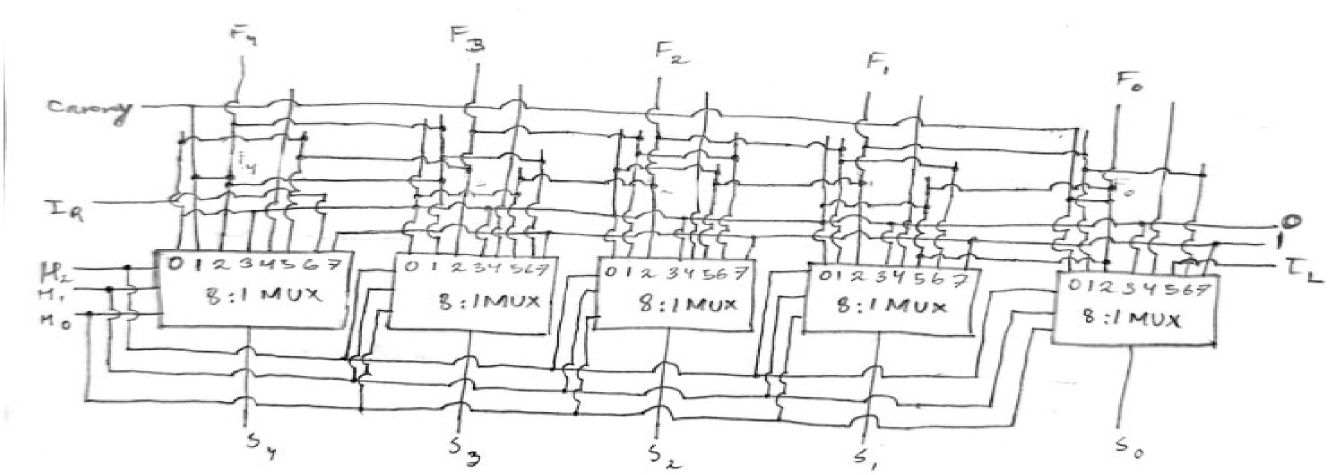
Table 2 Functions of control variables

Binary Code	Functions of selection variables					
	A	B	D	F with $C_{in} = 0$	F with $C_{in} = 1$	H
0 0 0	Input Data	Input Data	None	A-1	A, C←1	No Shift
0 0 1	R1	R1	R1	A+B	A+B+1	Shift Right with I_R=0
0 1 0	R2	R2	R2	A-B-1	A-B	Shift Left with I_L=0
0 1 1	R3	R3	R3	A, C←0	A+1	Circulate Left with Carry
1 0 0	R4	R4	R4	A′	A′	0's to the output Bus
1 0 1	R5	R5	R5	A ∩ B	A ∩ B	1's to the output Bus
1 1 0	R6	R6	R6	A XOR B	A XOR B	Circulate-Right with Carry
1 1 1	R7	R7	R7	A ∪ B	A ∪ B	Circulate-Right with Zero flag

Name: Mashkur Ibtesham Wazed
ID: 23-54266-3
Course: Microcontrollers and Embedded systems
Section: O

pq - a bcde - w
23 54266 3

③ Design a 5-bit shifter for the ~~following~~ shifting operations



4. Design an adder/subtractor circuit with one selection variable s and two inputs A and B: when s = 1 the circuit performs A+B. When *s* = 0 the circuit performs A-B by taking the 2's complement of B.

Name:- Mohammad Ansan Uddin

Id:- 22-47975-2

Course:- Microprocessor and Embedded

Sec: O

$\boxed{4}$

To design a 1-bit adder/ subtractor circuit using a single control input s and two 1-bit operands A and B:

when, s = 1, the circuit performs: A+B

s = 0, the circuit performs: A−B

using the 2's component method for subtraction

To implement both addition in a single circuit, we use the following approach:

- Subtraction:
  A − B = A + B + 1 using 2 components of B
- Addition:
  A + B

To accomodate both cases: modified B input:

$B' = B \oplus s$

if s = 1; $B' = B$

if s = 0; $B' = B$

Carry-in (cin):

$$C_{in} = \bar{S}$$

if S=1: Cin = 0

if S=0: Cin = 1 (to add 1 for 2's complement subtraction)

Using standard full-adder logic,

Sum:

$$Sum = A \oplus B \oplus S$$

Carry:

$$carry = A \cdot B' + (A \oplus B') \cdot C_{in}$$

where,

$$B' = B \oplus S$$

$$C_{in} = \bar{S}$$

Truth table

S	A	B	$\bar{S}$	$B' = B \oplus \bar{S}$	$C_{in} = \bar{S}$	Sum	Carry
0	0	0	1	1	1	0	1
0	0	1	1	0	1	1	0
0	1	0	1	1	1	1	1
0	1	1	1	0	1	0	1
1	0	0	0	0	0	0	0
1	0	1	0	1	0	1	0
1	1	0	0	0	0	1	0
1	1	1	0	1	0	0	1

Sum k-Map

$Sum = A \oplus B \oplus \bar{S}$

A\B	00	01	11	10
0	0	1	1	0
1	1	0	0	1

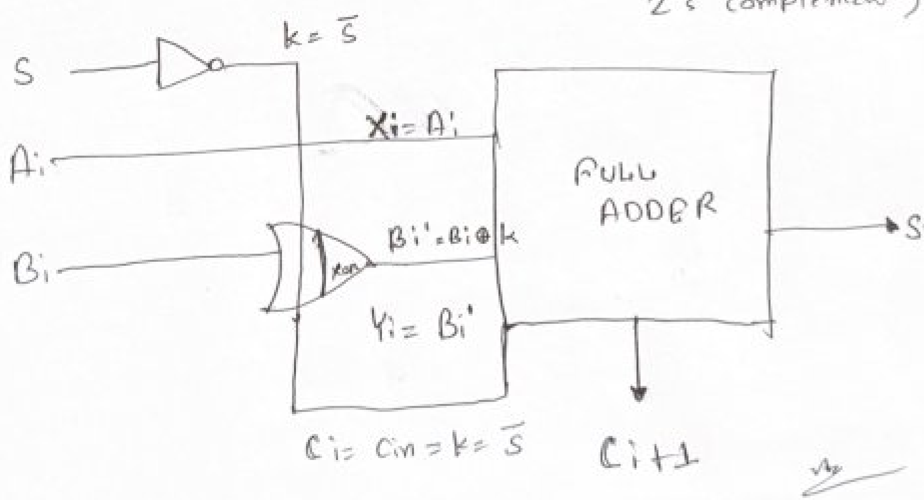
$\boxed{Sum = A \oplus B \oplus \bar{S}}$

Carry k-map

A\B	00	01	11	10
0	1	0	0	1
1	1	1	1	1

$\boxed{Carry = A \cdot B' + (A \oplus B') \cdot \bar{S}}$

Adder/Subtractor (s=1: A+B, s=0: A−B using 2's complement)



5. Write a 16-bit control word in binary and hexadecimal for the following micro-operations:
   a) Ra → Rr – Ra
   b) R1 → R2 XOR R4
   c) R5 → Rr+Ra
   d) R4 → R4 AND Rr
   e) R3 → SHL Rr

1 2 3	4 5 6	7 8 9	10 11 12	13	14 15 16
A	*B*	*D*	*F*	C_{in}	*H*

Binary code	Function of selection variables					
	A	*B*	*D*	*F* with $C_{in} = 0$	*F* with $C_{in} = 1$	*H*
0 0 0	Input data	Input data	None	$A, C \leftarrow 0$	$A + 1$	No shift
0 0 1	*R1*	*R1*	*R1*	$A + B$	$A + B + 1$	Shift-right, $I_R = 0$
0 1 0	*R2*	*R2*	*R2*	$A - B - 1$	$A - B$	Shift-left, $I_L = 0$
0 1 1	*R3*	*R3*	*R3*	$A - 1$	$A, C \leftarrow 1$	0's to output bus
1 0 0	*R4*	*R4*	*R4*	$A \vee B$	—	—
1 0 1	*R5*	*R5*	*R5*	$A \oplus B$	—	Circulate-right with *C*
1 1 0	*R6*	*R6*	*R6*	$A \wedge B$	—	Circulate-left with *C*
1 1 1	*R7*	*R7*	*R7*	$\bar{A}$	—	—

Ans. to the Que. No - 5

ID - 23 - 54270 - 3
P Q a b c d e - ~~R~~

a) Ra ← Rr − Ra
R5 ← R3 − R5

A	B	D	F	Cin	H	binary	Hexadecim
011	101	101	010	1	000 →	30376	76

b) R1 ← R2 XOR R4

A	B	D	F	Cin	H		
010	100	001	101	0	000	→ 20688	50D0

c) R5 ← Rr + Ra
R5 ← R3 + R5

A	B	D	F	Cin	H		
011	101	101	001	0	000	→ 30352	7690

d) R4 ← R4 AND Rr
R4 ← R4 AND R3

A	B	D	F	Cin	H		
100	011	100	100	0	000	→ 36416	8E40

e) R3 ← SHL Rr
R3 ← SHL R3

A	B	D	F	Cin	H		
011	011	011	100	0	010	→ 28098	6DC2

6. Draw the diagram of a 4-bit status register that has four status bits - carry, sign, zero and overflow flags. Find the four status bits performing the following operations of the two numbers of 8-bits:

   Number A: 100 + a + b

   Number B: 110 + r + a

   a) A + B
   b) A – B
   c) A XOR B
   d) A OR B
   e) A AND B

Solution:

Microprocessor Assignment:

Name: Chinmoy Gaten

ID: 22-48056-2

pq - abcde - r

∴ a=4, b=8 and r=2

(6)

V overflow	Z zero	S sign	C carry
Bit 3	Bit 2	Bit 1	Bit 0

Number A = 100 + a + b = 112

Number B = 110 + r + a = 110 + 2 + 4 = 116

∴ A < B. and 112 in Binary = 0111 0000

116 in Binary = 0111 0100

(a) A + B

```
  01110000
+ 01110100
----------
  11100100   -> 228 ∴ overflow.
```

∴ C = 0, S = 1, Z = 0, V = 1

(b) A - B

```
  01110000
- 01110100
----------
  11111100
```

∴ C = 0, S = 1, Z = 0, V = 0

(c) A XOR B

$$\begin{array}{r} 0111\,0000 \\ \oplus\ 0111\,0100 \\ \hline 0000\,0100 \end{array}$$

$C = 0$, $S = 0$, $Z = 0$, $V = 0$

(d) A OR B

$$\begin{array}{r} 0111\,0000 \\ |\ 0111\,0100 \\ \hline 0111\,0100 \end{array}$$

$C = 0$
$S = 0$
$Z = 0$
$V = 0$

(e) A AND B.

$$\begin{array}{r} 01110000 \\ \&\ 01110100 \\ \hline 01110000 \end{array}$$

$C = 0$, $S = 0$, $Z = 0$, $V = 0$